Takahiro Maeda

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Highlights

- First-authored papers at CVPR, ICCV, TMLR, IEEE RA-L, and Neurocomputing.
- Expertise in enterprise-tuned VLMs, flow models, and safety-critical prediction systems.
- Research internships at Carnegie Mellon University and Toyota; 1st place in the MPI-IS Real Robot Challenge 2020 (Phases 2 and 3).

Research Interests

Computer vision, robotics, human motion prediction, and multimodal models. Currently developing enterprise VLMs and AI agents at Toshiba, with long-term interests in robust perception for intelligent assistive robots.

Work Experience

Toshiba Corporation, Corporate Laboratory Apr 2025 – Present
AI Digital R&D Center, Collaborative AI Laboratory · Researcher (Machine Learning, VLMs, AI Agents), full-time

- **VLM fine-tuning and workflow design:** Fine-tuned VLMs for business processes that handle enterprise documents such as technical documents and drawings, and designed workflows that integrate these VLMs.
- **Knowledge transfer with LLMs/VLMs:** Researching methods that use LLMs/VLMs to extract and structure experts' work procedures and tacit knowledge into a knowledge base that AI agents can use.
- **Evaluation benchmark:** Built an initial internal benchmark for technical drawing interpretation to standardize evaluation of multimodal reasoning models.
- **Collaboration with business units:** Identifying workflow automation opportunities and defining requirements for future AI-agent integrations.

Toyota Technological Institute, Japan Apr 2021 – Sep 2021
Researcher

- Continued the MotionAug research started at TTIC (CVPR 2022).
- Continued this research in the doctoral program from October 2021.

Research Internships

Robotics Institute, Carnegie Mellon University, USA Sep 2024 – Dec 2024

- Fast human motion prediction with normalizing flows (CacheFlow; TMLR 2026).

Frontier Research Center, Toyota Motor Corporation, Japan May 2023 – Aug 2023

- Real-robot research on human mesh recovery in close proximity using the Toyota Human Support Robot (IEEE RA-L 2024).

Toyota Technological Institute at Chicago (TTIC), USA Sep 2019 – Dec 2019

- Started research on MotionAug, a physically corrected data augmentation method for human motion prediction (CVPR 2022).
- The stay led to a collaboration with M. Walter's group at TTIC on the Real Robot Challenge 2020 (1st place).

Education

Toyota Technological Institute, Japan Oct 2021 – Mar 2025
Doctoral Program, Information-aided Technology, Graduate School of Engineering
Completed course requirements; withdrew without degree

Toyota Technological Institute, Japan Apr 2019 – Mar 2021
Master of Engineering, Electronics and Information Engineering

Toyota Technological Institute, Japan Apr 2015 – Mar 2019
Bachelor of Engineering

Selected Publications

1. **T. Maeda**, J. Cao, N. Ukita, and K. Kitani. "CacheFlow: Fast Human Motion Prediction by Cached Normalizing Flow." *TMLR, 2026*.
2. **T. Maeda** and N. Ukita. "Continuously-indexed Flows for Fast Inference and Updating of Trajectory-prediction Densities." *Neurocomputing, 2026*.
3. H. Taketsugu, T. Oba, **T. Maeda**, S. Nobuhara, and N. Ukita. "Physical Plausibility-aware Trajectory Prediction via Locomotion Embodiment." *CVPR, 2025*.
4. **T. Maeda**, K. Takeshita, N. Ukita, and K. Tanaka. "Multimodal Active Measurement for Human Mesh Recovery in Close Proximity." *IEEE RA-L, 2024*.
5. **T. Maeda** and N. Ukita. "Fast Inference and Update of Probabilistic Density Estimation on Trajectory Prediction." *ICCV, 2023*.
6. **T. Maeda** and N. Ukita. "MotionAug: Augmentation with Physical Correction for Human Motion Prediction." *CVPR, 2022*.

Full publication list: Google Scholar (https://scholar.google.com/citations?user=oj90RrQAAAAJ)

Honors & Awards

- **Outstanding Reviewer**, NeurIPS 2026
- **Conference Presentation Award**, IEEE Japan Nagoya Section 2023
  For the presentation of MotionAug (CVPR 2022)
- **1st Place, Phases 2 and 3**, Real Robot Challenge 2020 2020
  Max Planck Institute for Intelligent Systems; prizes €3,500 and €9,000
- **Best Poster Award**, Machine Vision and Applications (MVA) 2019

Academic Service

- **Program Committee:** AAAI 2027
- **Conference reviewer:** CVPR, ICCV, NeurIPS, ICLR, ICRA, ACCV
- **Journal reviewer:** IEEE RA-L, IEEE TPAMI, CVIU

Skills

- **LLM/VLM training:** Domain adaptation (SFT, RLVR, OPSD)
- **LLM/VLM evaluation & ops:** Benchmark construction, prompt/context optimization; inference and deployment (vLLM, Ollama, Hugging Face)
- **AI agents:** ReAct, Tool Use; LangChain, Google ADK
- **AI coding:** Claude Code, Codex
- **Generative models:** Normalizing Flow, Diffusion, Flow Matching
- **Robotics:** ROS 2, Toyota Human Support Robot
- **Simulation:** Isaac Sim, Bullet Physics

Licenses & Languages

- **Certified Care Worker** (Japanese national license) Apr 2022
- **English:** TOEFL iBT 94 (R 29, L 24, S 18, W 23) Jul 2024